

FOCAL-VLA: Subtask-Guided Geometry Distillation and Implicit World Modeling for Vision–Language–Action Models

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Abstract—Vision–language–action (VLA) models built on pretrained vision–language models have demonstrated strong performance across diverse robotic manipulation tasks. However, VLA models that directly map current 2D observations to actions often lack sufficient spatial and temporal understanding, limiting their performance in precise and long-horizon manipulation. Recent methods enhance VLA models through geometric supervision and future-state prediction across the entire scene. However, these methods can suffer from redundant scene information, distracting the model from learning the geometry and dynamics relevant to the current interaction. To address this issue, we propose FOCAL-VLA, a framework that combines subtask-guided geometry distillation with implicit world modeling to learn representations of current spatial structure and future interaction dynamics. To focus geometric learning on the current subtask, we transfer geometric knowledge from VGGT to the VLA model by aligning geometry latents with features from subtask-relevant image regions. To capture the future 3D evolution of the current interaction, we incorporate implicit world modeling using Track4World features from current and future demonstration frames. The two complementary representations jointly guide action generation without running VGGT or Track4World at inference time. Experiments show that FOCAL-VLA outperforms baselines on both simulation benchmarks and real-world manipulation tasks. Project website: <https://zhiyuan-gao.github.io/FOCAL-VLA/>.

I. INTRODUCTION

Vision–language–action (VLA) models extend pretrained vision–language models to robotic control, enabling robots to perform diverse tasks and generalize to new objects and environments [1]–[4]. Advances in large-scale robot learning [5], [6] and training across embodiments [7] have further expanded these capabilities. However, precise and long-horizon manipulation requires more than semantic understanding: robots must perceive the spatial relationships between interacting entities and anticipate how an interaction will evolve [8]–[10]. To strengthen spatial understanding, recent methods incorporate 3D representations and spatial encodings into VLA models [11]–[13], or transfer geometric knowledge from pretrained 3D foundation models through representation alignment [8], [14], [15]. The latter provides geometric supervision during training without requiring the teacher at inference time. A complementary direction improves temporal understanding by incorporating motion history or

predicting future observations [16]–[18]. These approaches range from visual trace prompting [16] and joint image–action prediction [18] to learning future visual features, 3D point tracks, and scene flow [9], [19]–[21]. Together, these developments motivate combining representations of current spatial structure with those of future interaction dynamics.

The usefulness of spatial and temporal information, however, depends on its relevance to the current interaction. Recent studies address this requirement through visual attention conditioned on instructions and execution history [22], or through structured affordance representations and progress estimation [23]. For geometry and future-motion distillation, the spatial scope of the supervisory targets is therefore an important design choice. Existing methods that align all visual tokens [8], [14] or pool motion features across the entire scene [21] also supervise regions unrelated to the current interaction. Such supervision can introduce redundant scene information, distracting the model from learning the geometry and dynamics relevant to the current operation. Moreover, the information needed for action generation changes as the interaction progresses. For example, placing an object into a drawer requires understanding the object–drawer spatial relationship, whereas closing the drawer requires anticipating the drawer’s motion. These observations motivate constructing geometric and future-motion supervision around the entities involved in the current interaction.

To address these limitations, we propose FOCAL-VLA, built on $\pi_{0.5}$ [4], which combines subtask-guided geometry distillation with implicit world modeling to guide action generation (Fig. 1). To focus geometric learning on the current subtask, we distill features from frozen VGGT [24] into geometry latents by aggregating teacher features only from regions corresponding to the entities involved in that subtask. The selected regions change with the interacting entities across subtasks. To anticipate how the current interaction will unfold, we incorporate implicit world modeling through future motion latents learned from current observations. Frozen Track4World [25] extracts features from current and future demonstration frames. We aggregate features only from regions relevant to the current subtask to supervise the future motion latents. The two complementary representations jointly condition the action expert, without running either teacher at inference time. The subtask annotations and relevance masks used to construct distillation targets are not required at inference time.

We evaluate FOCAL-VLA on LIBERO [26], RoboCasa [27], and four real-world manipulation tasks. FOCAL-

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FOCAL-VLA

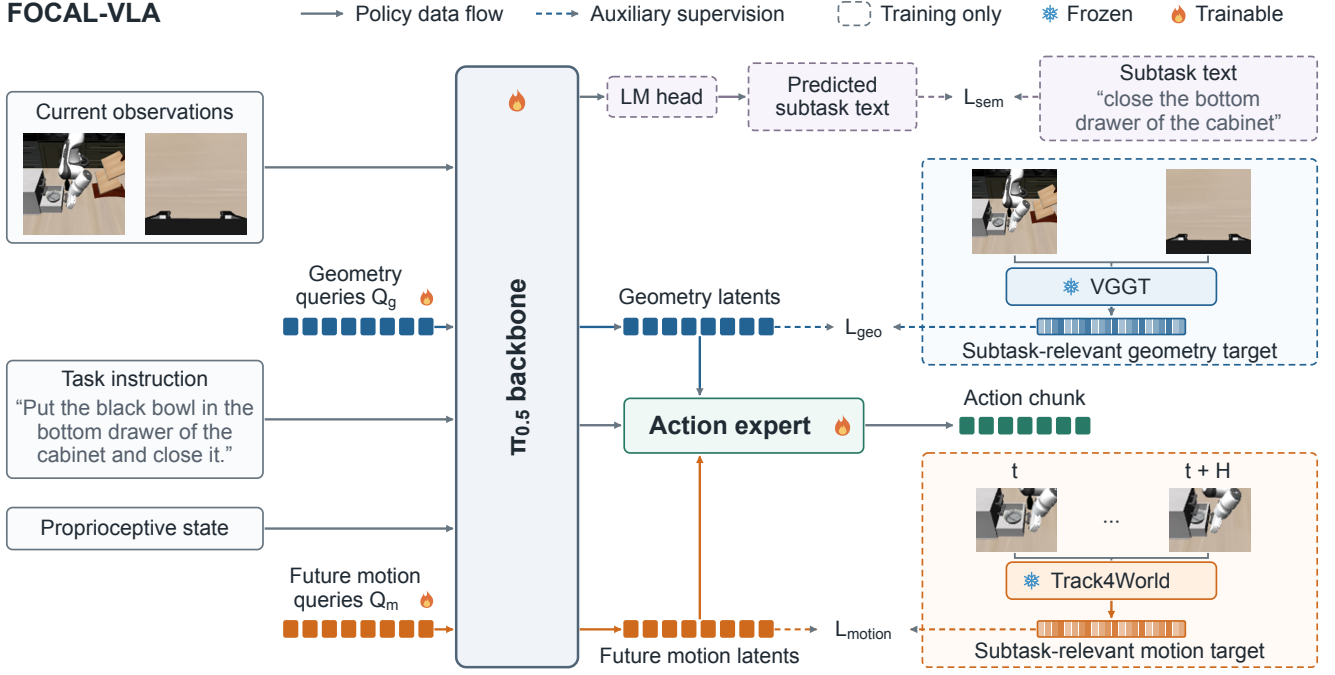


Fig. 1. Overview of FOCAL-VLA. The $\pi_{0.5}$ backbone processes current observations, the task instruction, proprioceptive state, and two groups of learnable queries. The output hidden states at the query positions form geometry and future motion latents, which jointly condition the action expert alongside the native vision–language context. Geometry distillation and implicit world modeling use cached, subtask-relevant targets constructed from frozen VGGT and Track4World features, respectively (Fig. 2). An auxiliary language-modeling branch predicts subtask text through the native LM head. Dashed borders indicate training-only branches; distillation prediction heads are omitted for clarity. The motion thumbnails show enlarged regions at t and $t + H$, where H is the action horizon; Track4World receives complete frames.

VLA performs competitively with existing VLA baselines on both simulation benchmarks and outperforms the $\pi_{0.5}$ baseline on real-world manipulation tasks. Ablation studies further demonstrate the complementary benefits of geometry and future motion supervision, while showing that restricting supervision to subtask-relevant regions outperforms full-image supervision.

In summary, our main contributions are:

- To focus geometric learning on the current subtask, we transfer VGGT’s geometric knowledge to the VLA model through distillation over subtask-relevant regions, adapting the selected regions as the interacting entities change.
- To anticipate how the current interaction will unfold, we introduce implicit world modeling through future motion latents, using future demonstration context while restricting spatial supervision to regions relevant to the current subtask.
- To evaluate the effectiveness of FOCAL-VLA, we conduct experiments on simulation benchmarks and real-world manipulation tasks, complemented by ablation studies examining the contributions of the distillation branches, supervision scope, and action conditioning.

II. RELATED WORK

A. Vision–Language–Action Models

VLA models adapt pretrained vision–language representations to robot action generation [1]–[3]. Representative designs include autoregressive action tokenization in OpenVLA [2] and continuous flow-matching action generation in π_0 [3]. Recent generalist models further expand capabilities for instruction following and transfer across tasks and embodiments: $\pi_{0.7}$ [28] uses diverse context conditioning to learn from heterogeneous data, while LingBot-VLA [29] scales pretraining across multiple dual-arm robot configurations. Our base policy, $\pi_{0.5}$ [4], combines a continuous action expert with heterogeneous co-training that includes semantic subtask prediction. FOCAL-VLA builds on this architecture and augments its policy representations through subtask-guided geometry distillation and implicit world modeling.

B. Geometry-Enhanced VLA Models

One approach to improving spatial understanding is to provide the policy with explicit geometric information, including point-cloud embeddings [30], [31], robot-centric pointmaps [32], and egocentric 3D position encodings [12]. Rather than supplying explicit geometric inputs to the policy, feature-distillation methods transfer geometric knowledge from pretrained 3D foundation models into policy representations without requiring the teacher at inference time. Spatial Forcing [8] and GLaD [14] align a single VLA backbone

layer with VGGT features, while ROCKET [33] extends alignment to multiple layers through a shared projector. 3DThinkVLA [15] incorporates both geometric and spatial reasoning features into action-query states to guide action prediction. These methods align geometric features across the entire image rather than focusing the alignment on regions relevant to the current subtask. We align the policy’s geometry latents with teacher features from subtask-relevant regions, adapting these regions as the interacting entities change across subtasks. The resulting geometry latents directly condition the action expert.

C. Future Prediction for VLA Models

One line of work guides VLA action generation by predicting future images or visual features. CoT-VLA [34] generates a future goal image before generating its action chunk, while DreamVLA [9] predicts future dynamic regions, depth, and semantic features. Related world-action models couple future video generation with action prediction or planning [35], [36].

Other methods represent future changes through motion features or 3D trajectories. CoWVLA [37] learns a factorized motion representation from video, while GeoPredict [38] combines future robot-track prediction with predictive 3D Gaussian geometry. Pri4R [19] uses auxiliary future 3D point-track prediction during training while retaining the original VLA architecture at deployment. LaMP [20] learns future 3D scene flow with a motion expert and supplies its denoising features to the action expert. Track4Action [21] pools scene and motion features from Track4World over an action-aligned clip into a tracker target that supervises query states. These query states condition action generation through gated fusion. Our method uses the same teacher but restricts feature distillation to regions relevant to the current subtask. This supervises the policy’s future motion latents using features that capture upcoming 3D changes in the interacting entities.

III. METHOD

In this section, we present FOCAL-VLA. We begin with the preliminaries (Sec. III-A), followed by subtask-guided geometry distillation (Sec. III-B) and implicit world modeling through future-informed motion distillation (Sec. III-C). We then describe action generation and training (Sec. III-D). An overview of FOCAL-VLA is shown in Fig. 1.

A. Preliminaries

We first review the VLA formulation using $\pi_{0.5}$ [4], which serves as our base policy. It combines a pretrained PaliGemma vision–language backbone with a continuous action expert. At time t , the backbone encodes the current multi-view observation o_t , task instruction ℓ , and tokenized proprioceptive state into context tokens C_t . Conditioned on C_t , the action expert predicts an action chunk $a_{t:t+H-1} = (a_t, \dots, a_{t+H-1})$, where a_t denotes a robot action and H is the prediction horizon. The action expert is trained on expert demonstrations through conditional flow matching and

generates actions by integrating a learned velocity field from Gaussian noise.

Building on this architecture, FOCAL-VLA introduces two groups of learnable queries, Q_g and Q_m , as additional input tokens to the vision–language backbone. Their output hidden states form the geometry latents Z_g and future motion latents Z_m , respectively, with the attention pattern defined in Sec. III-D. The geometry latents are aligned with subtask-relevant VGGT features, while the future motion latents are aligned with future-informed Track4World features. Both groups provide additional conditioning to the action expert alongside C_t .

B. Subtask-Guided Geometry Distillation

Existing geometry-distillation methods align VLA visual embeddings with 3D foundation-model features across the entire image. This alignment includes scene regions unrelated to the current operation and does not account for changes in the relevant entities across subtasks. To focus geometric learning on the current interaction, we propose subtask-guided geometry distillation, which aligns the policy’s geometry latents with features from our geometry teacher, VGGT [24], aggregated over entities involved in the current subtask (Fig. 2). The geometry query group Q_g consists of eight learnable tokens appended to C_t .

VGGT is a feed-forward multi-view model for camera and scene geometry estimation. The frozen model jointly processes the current-view images, yielding final-layer patch features $G_t^v \in \mathbb{R}^{P_g \times d_g}$ for each view $v \in \mathcal{V}$, where P_g is the number of patches per view and $d_g = 2048$. To select features relevant to the current subtask, we use an offline vision–language model to align manually specified subtask descriptions with demonstration frames and identify the entities involved. A binary relevance mask M_t^v is formed by taking the union of the whole-entity masks for entities involved in the current subtask. Let $w_{t,p}^v \in [0, 1]$ denote the coverage of the transformed relevance mask on patch p . We aggregate features only within these subtask-relevant regions, pooling within each view and averaging over views with nonzero mask coverage:

$$y_t^g = \frac{1}{|\mathcal{V}_t|} \sum_{v \in \mathcal{V}_t} \frac{\sum_{p=1}^{P_g} w_{t,p}^v G_{t,p}^v}{\sum_{p=1}^{P_g} w_{t,p}^v}, \quad (1)$$

where $\mathcal{V}_t = \{v : \sum_p w_{t,p}^v > 0\}$ is the set of valid views.

A prediction head applies per-query normalization, mean pooling, and a linear projection to produce $\hat{y}_t^g = h_g(Z_g) \in \mathbb{R}^{d_g}$. For each branch $b \in \{g, m\}$, we standardize targets and predictions using $\tilde{x} = (x - \mu_b)/\sigma_b$, where μ_b and σ_b are the per-dimension target mean and standard deviation computed on the training split. We align the geometry latents with the pooled VGGT target using a masked mean-squared error. Let $\mathcal{B}_g$ contain the batch samples with at least one valid view:

$$\mathcal{L}_{\text{geo}} = \frac{1}{|\mathcal{B}_g|d_g} \sum_{t \in \mathcal{B}_g} \left\| \tilde{\hat{y}}_t^g - \tilde{y}_t^g \right\|_2^2. \quad (2)$$

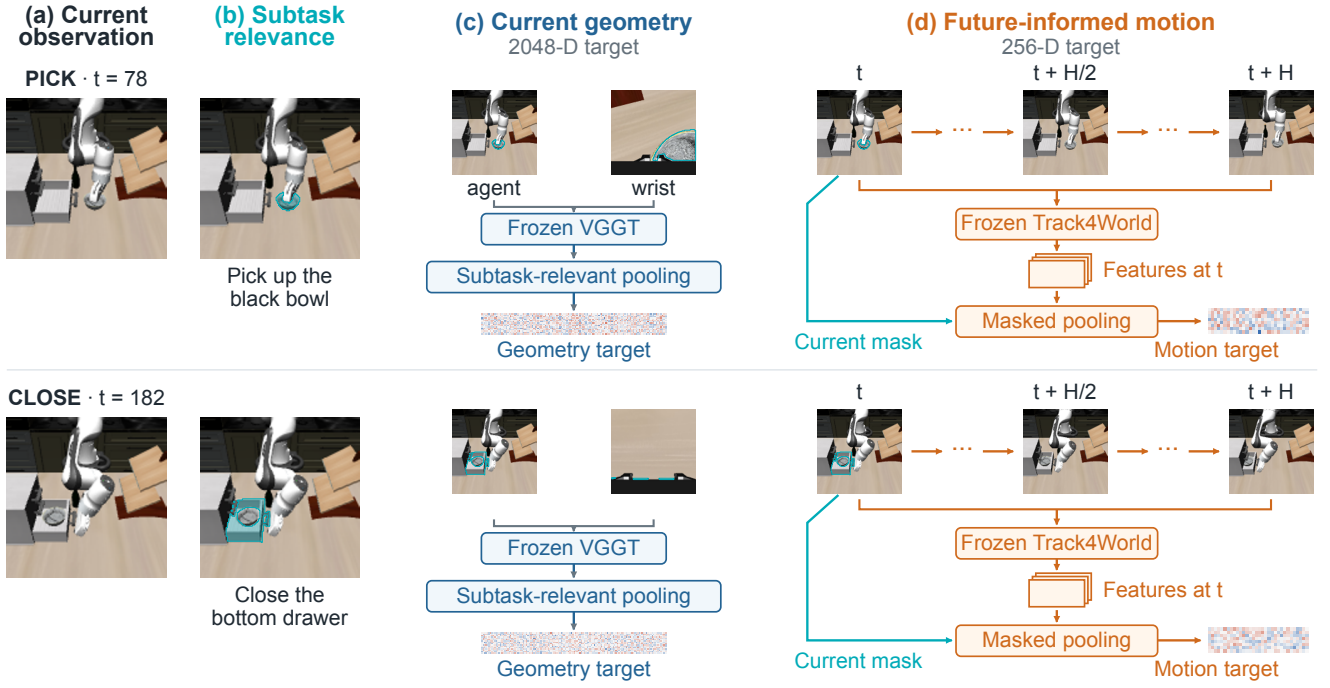


Fig. 2. Construction of subtask-relevant geometry and future motion targets. (a–b) PICK and CLOSE from the same demonstration select different entities: the black bowl and the bottom drawer, respectively. (c) VGGT features are pooled within the selected regions in each current view and averaged across valid views to form the geometry target. (d) Track4World processes current and future frames over the action horizon H . The current subtask mask selects features at time t , enriched by future context, for motion-target pooling. Mask overlays indicate pooling regions; both teachers receive complete images. Color blocks show standardized cached target vectors on a shared $[-4, 4]$ scale, not spatial maps; feature stacks are schematic.

C. Implicit World Modeling

We implement implicit world modeling through future-informed motion distillation. Conditioned on the current context C_t , the model learns to infer a latent representation of upcoming 3D interaction dynamics, using future demonstration frames only to construct training targets. Recent approaches incorporate future motion representations into VLA models. However, full-scene feature pooling [21] combines motion cues from interacting entities with features from regions unrelated to the current subtask. Our future-informed motion distillation addresses this issue by restricting feature aggregation to the currently relevant entities while retaining temporal context from future demonstration frames. Aggregated features from our motion teacher, Track4World [25], provide targets for the policy’s future motion latents (Fig. 2). We implement this branch with eight learnable query tokens Q_m appended to C_t .

Track4World is a feed-forward model that estimates dense 3D scene flow between video frames and reconstructs point trajectories in a shared world coordinate system. We use its frozen motion features, rather than the reconstructed trajectories, as distillation targets. By processing current and future demonstration frames together, the teacher provides motion features informed by the subsequent interaction. For a training anchor at time t , let $\mathcal{T}_t = \{\tau_k\}_{k=1}^K$ denote future timestamps sampled uniformly over the demonstration interval corresponding to the next action chunk, excluding the current frame. Sampling spans the full chunk interval without

truncation at subtask boundaries. The teacher processes the current agent-view frame o_t^{agent} together with $\{o_{\tau_k}^{\text{agent}}\}_{k=1}^K$, for $K + 1$ frames in total. The current agent-view mask M_t^{agent} selects the features used for distillation.

Let $D_{t,p} \in \mathbb{R}^{d_m}$ denote the final output of Track4World’s temporally global 3D flow-feature aggregator at source time index zero and source patch p , after the teacher has processed the full clip, where $d_m = 256$. Mapping the current relevance mask to the teacher’s 45×45 source grid ($P_m = 2025$) gives coverage weights $u_{t,p}$. The motion target is

$$y_t^m = \frac{\sum_{p=1}^{P_m} u_{t,p} D_{t,p}}{\sum_{p=1}^{P_m} u_{t,p}}. \quad (3)$$

Each $D_{t,p}$ incorporates temporal context from the full clip, while the current mask determines which entities contribute to y_t^m . Aligning Z_m with this target trains the policy to infer future motion features of the current subtask’s entities from its current context. Future evolution is thus modeled in latent feature space, without explicitly reconstructing future images or 3D trajectories.

A separate head with the same normalize–mean–project structure predicts $\hat{y}_t^m = h_m(Z_m) \in \mathbb{R}^{256}$, supervised by a masked Smooth L1 loss with transition parameter $\beta = 1$ in the standardized feature space:

$$\mathcal{L}_{\text{motion}} = \frac{1}{|\mathcal{B}_m|} \sum_{t \in \mathcal{B}_m} \text{SmoothL1}_{\beta=1}(\hat{y}_t^m, y_t^m). \quad (4)$$

Here Smooth L1 is averaged over the feature dimensions,

and $\mathcal{B}_m$ contains batch samples with a valid current agent mask and all K sampled future frames.

D. Action Generation and Training

The geometry and future motion query groups attend independently to C_t and to tokens within their own groups. The action expert attends to both groups alongside C_t , while the context tokens retain their native pathway. The attention pattern is

$$\begin{aligned} C_t &\leftarrow C_t, \\ Z_g &\leftarrow C_t + Z_g, \\ Z_m &\leftarrow C_t + Z_m, \\ \text{Action} &\leftarrow C_t + Z_g + Z_m. \end{aligned} \quad (5)$$

Here each left-hand side denotes the token group being updated, the right-hand side lists the groups it may attend to, and $+$ denotes their union. For the action expert, only external conditioning groups are shown; its native internal attention pattern is retained.

To provide the shared context with subtask-level semantic guidance, we also train the model’s native language-modeling branch to predict the subtask text obtained in Sec. III-B. This branch reuses the PaliGemma tokenizer, token embeddings, transformer, and tied language-model head to generate the text autoregressively from C_t . Given target tokens $s_{1:L_s}$ of length L_s , the semantic objective is

$$\mathcal{L}_{\text{sem}} = -\frac{1}{L_s} \sum_{k=1}^{L_s} \log p_{\theta}(s_k | C_t, s_{<k}), \quad (6)$$

including the end-of-sequence token and excluding padded positions. The native autoregressive branch requires no additional semantic query or semantic embedding head. Its attention mask allows target tokens to attend to C_t and their causal prefix, but prevents the context, geometry, future motion, and action tokens from attending to semantic target tokens. The semantic loss therefore trains the shared vision–language context without providing ground-truth subtask text to the action pathway.

The overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_g \mathcal{L}_{\text{geo}} + \lambda_s \mathcal{L}_{\text{sem}} + \lambda_m \mathcal{L}_{\text{motion}}, \quad (7)$$

where $\mathcal{L}_{\text{action}}$ is the native $\pi_{0.5}$ action objective, and λ_g , λ_s , and λ_m weight the three auxiliary objectives. We jointly optimize the policy parameters, query embeddings, and prediction heads using the cached distillation targets.

At inference time, the policy computes Z_g and Z_m from the current observation, instruction, and proprioceptive state and uses them to condition the action expert. The teachers, future demonstration frames, relevance masks, and semantic targets are not used.

IV. EXPERIMENTS

We evaluate FOCAL-VLA on the LIBERO and RoboCasa simulation benchmarks and on real-world robotic manipulation tasks. We further conduct ablation studies to examine the contributions of individual components and the effects of our design choices.

TABLE I
COMPARISON ON LIBERO (SUCCESS RATE, %). BOLD INDICATES THE BEST RESULT IN EACH COLUMN.

Method	Spatial	Object	Goal	Long	Avg.
OpenVLA [2]	84.7	88.4	79.2	53.7	76.5
π_0 [3]	96.8	98.8	95.8	85.2	94.2
$\pi_{0.5}$ [4]	98.8	98.2	98.0	92.4	96.9
OpenVLA-OFT [39]	97.6	98.4	97.9	94.5	97.1
GR00T N1.7 [40]	95.0	100.0	98.0	93.0	96.5
SpatialVLA [12]	88.2	89.9	78.6	55.5	78.1
DreamVLA [9]	97.5	94.0	89.5	89.5	92.6
GeoPredict [38]	98.0	98.2	95.7	94.0	96.5
Track4Action [21]	95.0	99.6	97.6	95.8	97.0
FOCAL-VLA	100.0	98.6	97.8	95.0	97.9

A. Experimental Setup

a) Simulation benchmarks: We train one policy per benchmark. LIBERO [26] comprises Spatial, Object, Goal, and Long (LIBERO-10), each with ten tasks; the policy uses agent and wrist views and predicts and executes 10-step action chunks. For RoboCasa [27], we use Atomic-24 under Human-50 (50 demonstrations per task), with left agent, wrist, and right agent views; the policy predicts 50-step chunks and executes 25 steps before replanning. We evaluate 50 episodes per task, reporting LIBERO’s per-suite success rates and their unweighted average, and RoboCasa’s unweighted average task success rate.

b) Real-world experiments: We use a Franka Emika Panda with one fixed external Orbbec Astra Pro RGB camera (Fig. 3). Fig. 4 illustrates four tasks: sequentially placing a banana and mandarin into instructed plates (possibly the same plate), fully opening a specified upper or lower drawer, storing a banana in a specified initially open upper or lower drawer and closing it, and closing the microwave. We collect 50–150 successful demonstrations per task using Space-Mouse teleoperation, totaling 350 demonstrations recorded at 10Hz with 640×480 RGB observations. FOCAL-VLA and $\pi_{0.5}$ each learn one policy across all four tasks, predicting and executing 10-step chunks. We report per-task and overall success rates over 20 trials per task, using the same initial-layout distribution for both methods. Success requires completing all instructed actions in order and reaching the specified final configuration.

c) Implementation details: We fine-tune the official OpenPI $\pi_{0.5}$ implementation end to end. Offline annotation (Sec. III-B) uses Qwen3-VL-8B-Instruct [41] to identify subtask starts and entities from sampled demonstration frames, task instructions, and manually specified subtask descriptions. Entities include manipulated objects and associated containers, supports, or articulated objects, excluding the gripper. Masks come from simulator annotations or entity-prompted Grounded-SAM-2 [42], [43] for real-world data. Frozen VGGT and Track4World targets are cached offline; RoboCasa motion targets use the current frame and ten future frames uniformly sampled over the 50-step chunk.

Training uses eight NVIDIA A100 80GB GPUs and

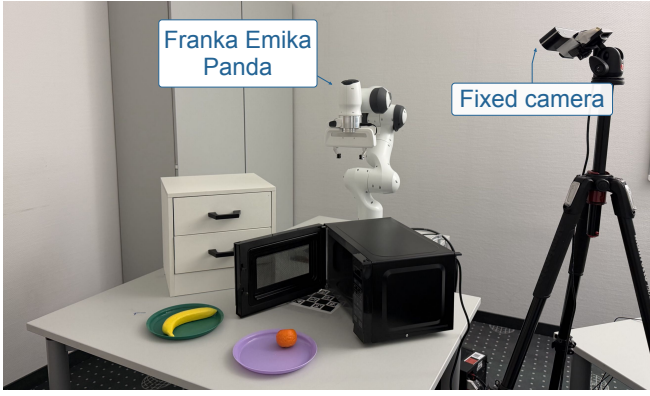


Fig. 3. Real-world platform with a Franka Emika Panda and a fixed external camera. The workspace contains the objects used across the four manipulation tasks.

AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$, weight decay 10^{-10} , gradient-norm clipping at 1.0). Simulation training uses 30,000 updates with global batch sizes of 256 for LIBERO and 128 for RoboCasa; real-world training uses 16,000 updates with a batch size of 256. We linearly warm up to 5×10^{-5} over 10,000 updates in simulation and 5,200 in real-world experiments. The learning rate then remains constant, except on RoboCasa, where it decays to 5×10^{-6} with a cosine schedule. For the full model, we use $\lambda_g = 0.05$, $\lambda_s = 0.01$, and $\lambda_m = 0.05$, with ablation-specific changes described below. We use the official $\pi_{0.5}$ training and evaluation seeds.

d) Baselines: We evaluate $\pi_{0.5}$ on RoboCasa using the same training and evaluation settings as FOCAL-VLA. Results for the other simulation baselines are taken from the cited sources. GR00T N1.7 results follow the preliminary official LeRobot evaluation [40]. RoboCasa results for KYC, GeoVLA, and PointVLA use Pointmap’s $\pi_{0.5}$ -based reimplementations [32]; Being-H0.5 uses its RoboCasa-specific model.

B. Evaluation Results

a) LIBERO: Table I summarizes results on LIBERO. FOCAL-VLA achieves an average success rate of 97.9%, improving over $\pi_{0.5}$ by 1.0 percentage point. It reaches 100.0% on Spatial and obtains its largest gain on Long, improving from 92.4% to 95.0%. These results are consistent with our joint modeling of current geometry and future interaction dynamics: geometry distillation emphasizes the spatial relationships needed for the current operation, while implicit world modeling complements this information with anticipated changes in the interacting entities. As a task progresses, the supervision follows the entities involved in each subtask rather than treating all image regions equally.

b) RoboCasa: Table II reports results on RoboCasa. FOCAL-VLA achieves 63.4% average success, improving over $\pi_{0.5}$ by 8.2 percentage points and outperforming the other listed methods. Kitchen manipulation requires selecting the objects and fixtures involved in the current operation from a cluttered scene. Subtask-guided geometry distillation

TABLE II
COMPARISON ON ROBOCASA. THE BEST SUCCESS RATE IS SHOWN IN BOLD.

Method	Avg. SR (%)
Base VLAs	
π_0 [19]	38.8
$\pi_{0.5}$ [4]	55.2
OpenVLA-OFT [19]	33.1
Being-H0.5 [44]	53.9
3D inputs required at inference	
KYC [32], [45]	59.1
GeoVLA [31], [32]	57.1
PointVLA [30], [32]	57.3
Robot-Centric Pointmap [32]	62.9
3D inputs not required at inference	
GeoPredict [38]	52.4
GaussianDream [46]	54.8
Pri4R [19]	57.0
FOCAL-VLA	63.4

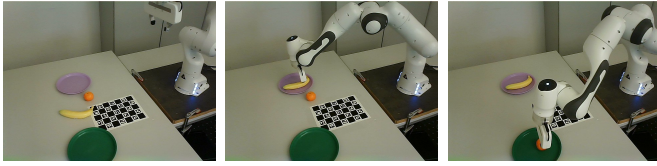
targets the spatial structure of these entities, while implicit world modeling captures their future interaction dynamics using features from current and future demonstration frames. The two branches therefore provide complementary information about the same interaction: its current configuration and its future evolution. This design offers an explanation for the improvement over the base policy, with the ablations below examining the roles of both distillation branches, supervision scope, and action conditioning.

c) Real-world experiments: Table III reports the real-world results on the four tasks illustrated in Fig. 4. FOCAL-VLA improves success on all four tasks, raising the overall success rate from 47.5% to 63.8%. Two-fruit rearrangement shows the largest gain and requires switching between instructed objects and destinations. Its improvement provides further evidence of the method’s effectiveness in sequential manipulation. Drawer storage similarly requires the robot to place the banana inside the drawer and then switch to closing it. Completing this sequence involves a change in both the interacting entity and the required motion. The gain is consistent with our motivation to associate geometric and future motion information with the current subtask: the policy must use the banana-drawer spatial relationship during placement and the drawer’s subsequent motion during closing. Gains on Drawer opening and Close the microwave indicate that the benefits also extend to individual articulated-object interactions, rather than being limited to tasks with multiple subtasks. Some real-world grasping failures occur when the target object is occluded in the single fixed camera view.

C. Ablation Studies

We ablate the two distillation branches, supervision scope, and action conditioning on RoboCasa under the same training and evaluation protocol (Table IV).

a) Geometry and future motion: To evaluate each teacher’s contribution, we separately remove the geometry



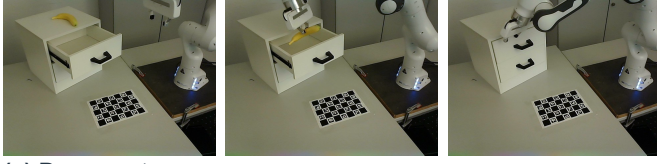
(a) Two-fruit rearrangement

Place the banana and mandarin into green or purple plates in the instructed order.



(b) Drawer opening

Open the upper or lower drawer.



(c) Drawer storage

Put the banana into the upper or lower drawer, then close it.



(d) Close the microwave

Close the microwave door.

Fig. 4. Illustration of the four real-world tasks. Representative frames from each task are shown in temporal order from left to right, viewed from the fixed external policy camera. Task descriptions appear below each sequence. A uniform color correction is applied for display.

TABLE III

REAL-WORLD TASK SUCCESS RATES (%). EACH METHOD IS EVALUATED IN 20 TRIALS PER TASK.

Task	$\pi_{0.5}$	FOCAL-VLA
Two-fruit rearrangement	25.0	50.0
Drawer opening	70.0	80.0
Drawer storage	20.0	40.0
Close the microwave	75.0	85.0
Overall	47.5	63.8

TABLE IV

ABLATIONS ON ROBOCASA.

Variant	Success (%)
Full method	63.4
Without geometry	60.0
Without future motion	59.6
Task-level entity supervision	61.9
Full-image supervision	58.7
Supervision-only	58.5

or future motion queries together with the corresponding distillation loss. Removing geometry reduces success from 63.4% to 60.0%, while removing future motion reduces it to 59.6%. The drop without future motion supports learning anticipated interaction changes in addition to current geometry. Conversely, the drop without geometry shows the benefit of retaining a dedicated representation of the current spatial configuration. Together, these results support combining subtask-relevant geometry distillation with implicit world modeling, rather than relying on either learning signal alone.

b) Supervision scope: To evaluate the scope of supervision, we compare subtask-relevant supervision with task-level entity supervision and full-image supervision. Different subtasks within a task may involve different entities. Task-level entity supervision uses the union of entities relevant

to all subtasks throughout the task, whereas subtask-level supervision selects only those relevant to the current subtask. In both cases, masks follow the entities in each frame. Both geometry and future motion distillation use these masks, with all other settings unchanged. Full-image supervision instead pools VGGT and Track4World features over the entire image [8], [14], [21]. Task-level entity supervision achieves 61.9% success, exceeding full-image supervision at 58.7%, while the full model achieves 63.4%. These results suggest that selecting task-relevant entities improves performance, with a further 1.5-percentage-point gain from adapting the supervised entities to the current subtask.

c) Action conditioning: To evaluate the benefit of direct action conditioning, we retain both distillation branches but disconnect their output latents from the action expert, yielding a supervision-only variant. The supervision-only variant achieves 58.5% success, compared with 63.4% for the full model. This 4.9-percentage-point gain supports directly conditioning the action expert on the learned latents, rather than using them solely as auxiliary training signals.

V. LIMITATIONS

FOCAL-VLA relies on offline subtask annotation and teacher-target construction, which add data-preparation and computational costs. Errors in subtask boundaries or entity masks can affect the quality of the distillation targets. Our evaluation covers two simulation benchmarks and a single real-world robot platform, leaving generalization to other embodiments and task families unexplored. Future work will investigate reducing the annotation effort and extending evaluation to more diverse robotic settings.

VI. CONCLUSION

We presented FOCAL-VLA, a framework for enhancing VLA models through subtask-guided geometry distillation and implicit world modeling. Geometry distillation aligns the policy’s geometry latents with VGGT features from entities relevant to each subtask. Implicit world modeling

is implemented through future-informed motion distillation, using Track4World features from current and future demonstration frames. These features are pooled over subtask-relevant regions to construct the distillation targets. The two groups of latents jointly guide action generation without requiring teacher models or subtask annotations at inference time. Experiments on LIBERO, RoboCasa, and real-world manipulation tasks show higher average success rates than the $\pi_{0.5}$ baseline. These results highlight the value of subtask-relevant geometric and future motion knowledge for improving VLA manipulation performance.

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